

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

Department of Electrical Engineering

VIBRATION-BASED BEARING FAULT CLASSIFICATION

USING SIGNAL PROCESSING AND MACHINE LEARNING

Project Report

Submitted by

Rohan Meena

Electrical Engineering

Abstract

Rolling element bearings are critical components in rotating machinery, and localized defects can produce characteristic changes in the vibration response of the machine. Early identification of the fault location is therefore important for condition monitoring and predictive maintenance.

This project develops a vibration-based bearing fault classification framework combining digital signal processing and classical machine learning. CWRU-derived vibration recordings are processed using multiband filtering, Hilbert envelope analysis and Welch power spectral density estimation. Physically motivated features are extracted around characteristic fault-related frequency regions. Window-level features are subsequently aggregated at the recording level using the median and interquartile range (IQR), producing a compact representation for machine learning.

The final classification problem consists of three fault classes: ball fault, inner-race fault and outer-race fault. Five classical classifiers are compared using Leave-One-Group-Out cross-validation, where recordings associated with the same physical fault instance are kept within the same validation group. This reduces the risk of information leakage between different operating conditions of the same physical defect.

The final Extra Trees model using median and IQR features achieves 83.33% pooled grouped cross-validation accuracy, 82.74% balanced accuracy and 82.0% macro-F1 across 60 recordings belonging to 15 physical fault groups. The results demonstrate the usefulness of combining physics-motivated vibration features with machine learning for bearing fault classification, while also highlighting limitations related to dataset size, feature selection, class imbalance and cross-domain generalization.

Keywords: Bearing fault diagnosis, vibration analysis, Hilbert transform, envelope analysis, feature engineering, Extra Trees, machine learning, condition monitoring.

1. Introduction

1.1 Background

Rolling element bearings are widely used in rotating machinery to support shafts and reduce friction between moving components. They are found in electric motors, pumps, turbines, gearboxes and many other industrial systems. Because bearings operate continuously under mechanical loading, defects can develop in the inner race, outer race or rolling elements.

A localized defect can generate repeated mechanical impacts as the bearing rotates. These impacts excite structural resonances and modify the vibration signature of the machine. Consequently, vibration measurements can provide useful information about the condition and location of a bearing defect.

1.2 Motivation

Unexpected bearing failure can result in machine downtime, maintenance costs and interruption of industrial processes. Conventional inspection methods may require the machine to be stopped or the bearing to be physically inspected. Vibration-based condition monitoring provides a non-invasive approach in which the machine condition can be inferred from measured vibration signals.

However, vibration signals are often complex and contain information from multiple mechanical sources. Direct inspection of a time-domain waveform may not clearly distinguish different fault locations. Signal processing can therefore be used to isolate frequency regions and extract features related to repetitive fault impacts. Machine learning can then learn the mapping between these features and known fault categories.

1.3 Problem Statement

The objective of this project is to develop a vibration-based supervised machine learning framework capable of classifying localized bearing faults into three categories:

1. Ball fault,
2. Inner-race fault,
3. Outer-race fault.

The input to the system is a vibration recording together with rotational speed information. The output is the predicted fault category.

1.4 Objectives

The major objectives of the project are:

1. Study the vibration characteristics associated with different bearing fault locations.
2. Develop a signal-processing pipeline for extracting meaningful vibration features.
3. Incorporate characteristic fault-frequency information into the feature representation.
4. Aggregate window-level features into recording-level descriptors.
5. Compare classical machine learning classifiers under group-aware validation.
6. Analyze the strengths and limitations of the resulting model.

1.5 Scope

The final model performs three-class fault classification. A healthy or normal-bearing class is not included in the final training problem. The work focuses on CWRU-derived vibration recordings and does not claim universal generalization to arbitrary machines or bearing types.

2. Background and Theoretical Foundation

2.1 Rolling Element Bearing Faults

A rolling element bearing consists primarily of an inner race, outer race, rolling elements and a cage. Defects occurring at different locations produce different periodic impact patterns.

The principal fault categories considered in this project are ball, inner-race and outer-race faults. Their characteristic frequencies depend on bearing geometry and shaft rotational speed.

2.2 Characteristic Fault Frequencies

The shaft rotational frequency is calculated from the measured rotational speed as

$$f_s = \frac{\mathrm{RPM}}{60}. \tag{1}$$

The project uses characteristic multipliers for the major fault-frequency families. For the implemented CWRU-specific feature extractor, the multipliers are:

Table 1: Characteristic frequency multipliers used in the feature extractor.

Frequency family	Multiplier
Inner race	5.4152
Outer race	3.5848
Cage	0.39828
Rolling element	4.7135

For harmonic number h, the target frequency is calculated as

$$f_{\text{target}} = M f_s h, \qquad h \in \{1, 2, 3\}, \tag{2}$$

where M is the characteristic multiplier.

2.3 Envelope Analysis

Bearing defects can generate repeated impacts that excite higher-frequency structural resonances. Envelope analysis is useful because it extracts amplitude modulation associated with these repetitive impacts.

For a vibration signal $x(t)$, the analytic signal is

$$x_a(t) = x(t) + j\mathcal{H}\{x(t)\}, \tag{3}$$

where $\mathcal{H}\{\cdot\}$ denotes the Hilbert transform.

The envelope is then

$$E(t) = |x_a(t)|. \tag{4}$$

The envelope is subsequently analyzed in the frequency domain to identify power concentrated near expected fault-related frequencies.

3. Dataset and Problem Formulation

3.1 Dataset

The project uses CWRU-derived MATLAB vibration recordings. The final recording-level development table contains 60 recordings belonging to 15 physical fault-instance groups.

Rotational Speed Information Rotational speed is an important input to the feature extraction process because the expected bearing fault frequencies depend on the shaft speed. The rotational speed in revolutions per minute (RPM) is converted into shaft frequency as

$$f_s = \frac{\text{RPM}}{60}, \tag{5}$$

where f_s is the shaft frequency in hertz.

The feature extraction code normally reads the RPM value from the MATLAB recording. However, for some additional CWRU recordings where this information is unavailable, approximate nominal speeds are assigned based on the load condition.

Table 2: Approximate RPM values used when recording metadata is unavailable.

Load Index	Approximate RPM
0	1797
1	1772
2	1750
3	1730

These fallback values are approximate rather than directly measured speeds. For example, the RPM value read from the selected recording B007_0.mat was 1796 RPM.

Table 3: Final dataset composition.

Property	Value
Recording-level samples	60
Physical fault groups	15
Number of fault classes	3
Base features selected	76
Median + IQR model inputs	152

3.2 Class Distribution

Table 4: Distribution of fault classes.

Fault class	Recordings
Ball fault	16
Inner-race fault	16
Outer-race fault	28
Total	60

3.3 Defect Size Distribution

The recordings contain different defect sizes:

Table 5: Distribution of recordings by defect diameter.

Defect diameter	Recordings
0.007 in	20
0.014 in	12
0.021 in	20
0.028 in	8

The 0.028-inch subset does not contain outer-race examples, and therefore cannot provide a balanced three-class comparison at that defect size.

3.4 Problem Formulation

Each vibration recording is treated as one supervised classification instance after window-level features are aggregated. The target variable represents the fault category:

$$y \in \{\text{ball}, \text{inner-race}, \text{outer-race}\}. \tag{6}$$

The final feature representation contains 76 selected base features. For each base feature, the median and IQR across signal windows are retained:

$$76 + 76 = 152 \tag{7}$$

model inputs.

3.5 Overall Pipeline

The project follows a sequence of signal processing and machine learning steps to identify the type of bearing fault from vibration recordings. The complete process is described below.

1. Load the MATLAB vibration recording and obtain its rotational speed (RPM).
2. Apply the Butterworth filter bank to the complete vibration signal to obtain different frequency bands.
3. Calculate the Hilbert envelope of each filtered signal to capture the amplitude variations caused by repeated impacts.
4. Divide the filtered signals and their corresponding envelopes into 0.5-second windows.
5. Extract statistical and frequency-domain features from each window using Welch's power spectral density estimation.
6. Combine the window-level features using the median and interquartile range (IQR) to obtain one feature representation for each recording.
7. Apply preprocessing steps, including missing-value imputation, variance filtering, correlation filtering and feature selection.
8. Train and compare five classical machine learning classifiers.
9. Evaluate the classifiers using Leave-One-Group-Out cross-validation, keeping recordings from the same physical fault instance together.
10. Train the selected Extra Trees classifier using the complete development dataset and save the trained model.

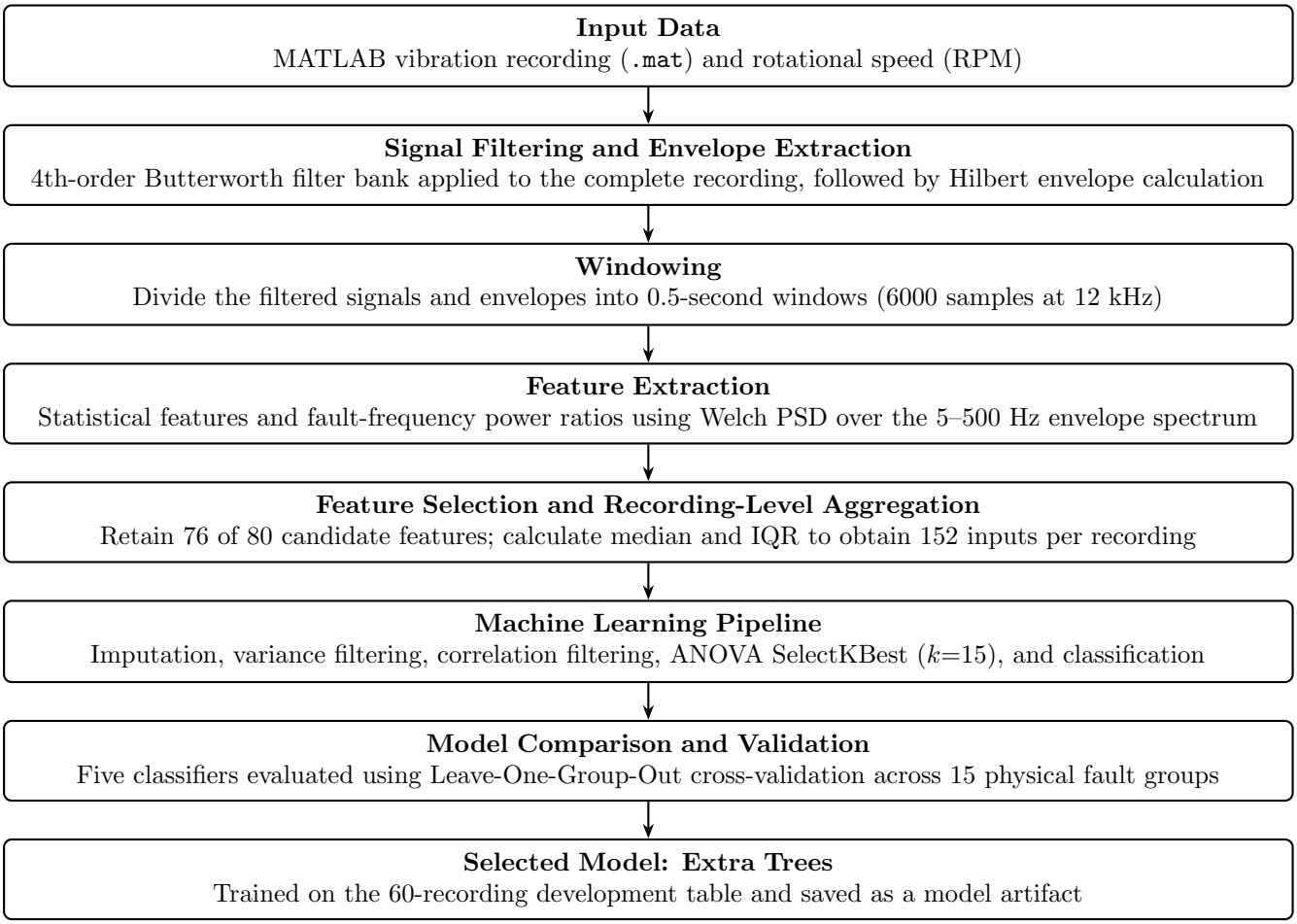


Figure 1: Overall workflow implemented for bearing fault classification.

3.6 Signal Windowing

The vibration signal is divided into 0.5-second windows. At a sampling frequency of 12 kHz, each window contains

$$N = 12000 \times 0.5 = 6000 \tag{8}$$

samples.

Windowing enables the extraction of multiple local feature vectors before recording-level statistical aggregation.

3.7 Multiband Filtering

The signal is processed through multiple frequency bands using Butterworth bandpass filters.

Table 6: Frequency bands used during feature extraction.

Band	Frequency range
Band 1	1000–2000 Hz
Band 2	2000–3000 Hz
Band 3	3000–4000 Hz
Band 4	4000–5000 Hz
Baseline	2000–5000 Hz

The filters used in this project are fourth-order Butterworth bandpass filters implemented in second-order-section (SOS) form. Forward-backward filtering is performed using the `sosfiltfilt` function. This avoids introducing a net phase shift in the filtered signal.

An important part of the implementation is that filtering is applied to the complete recording before it is divided into 0.5-second windows. The Hilbert envelope is also calculated from the complete filtered signal. The resulting filtered signals and envelopes are then divided into windows for feature extraction.

The purpose of multiband processing is to capture fault-related information appearing in different resonance regions.

3.8 Envelope and Spectral Features

For each filtered signal, the Hilbert envelope is calculated. Welch's method is then used to estimate the envelope power spectral density.

The total envelope power in the 5–500 Hz range is used as a reference. Power around expected fault-frequency targets is calculated relative to this total power.

The resulting ratio is

$$R = \frac{P_{\text{local}}}{P_{\text{total}}}, \tag{9}$$

where P_{local} represents power in a narrow region around the target fault frequency.

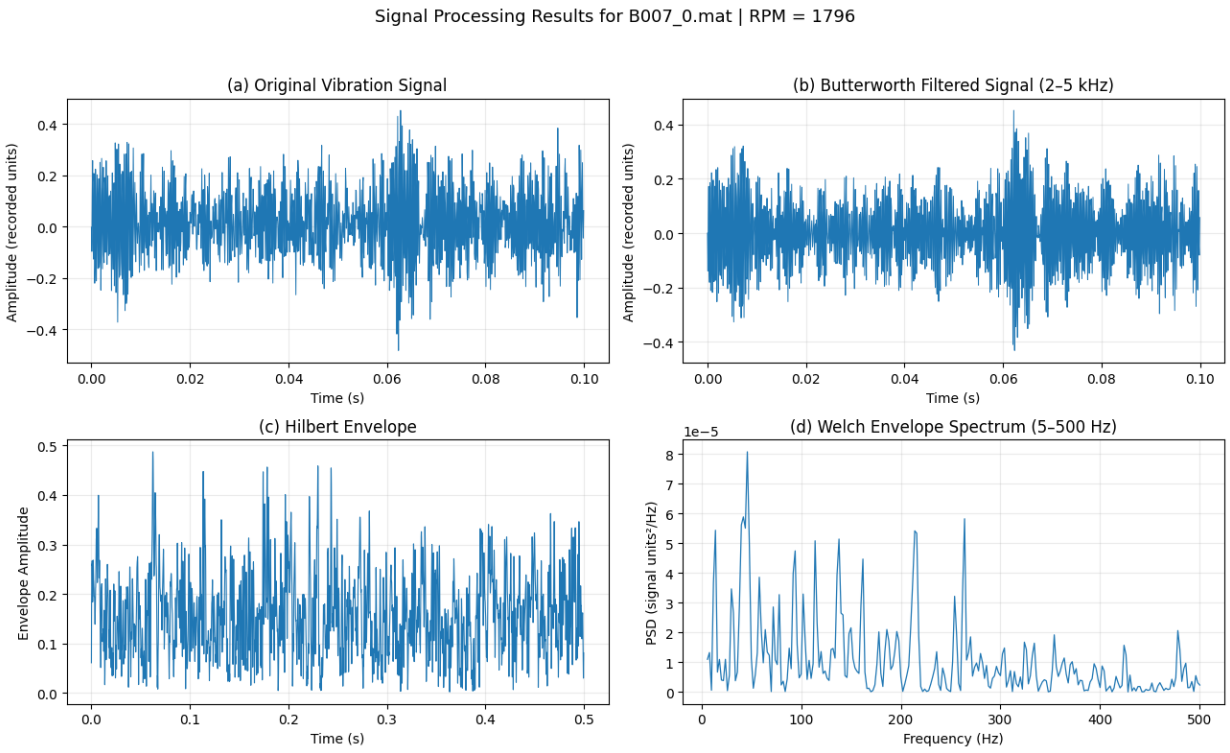


Figure 2: Signal processing stages for the CWRU bearing recording B007_0.mat. The figure shows the original vibration signal, the Butterworth-filtered signal in the 2–5 kHz band, its Hilbert envelope, and the corresponding Welch envelope spectrum from 5–500 Hz.

3.9 Feature Extraction

The extractor generates features representing signal magnitude, envelope statistics and fault-frequency power concentration.

The major feature types are:

- Bandpass RMS,
- Envelope standard deviation,
- Envelope kurtosis,
- Envelope spectral power,
- Fault-family harmonic power ratios.

The feature extractor is designed to calculate 80 candidate features from each signal window. However, the final notebook uses a prefix-based selection rule that retains 76 of these features. Four baseline features are excluded by this selection rule. Therefore, the feature representation used for model development contains 76 features per window.

These selected features are later summarized at the recording level using their median and interquartile range. This produces 152 inputs for each recording: 76 median values and 76 IQR values.

3.10 Recording-Level Aggregation

Since each recording produces multiple windows, directly treating every window as an independent sample could introduce undesirable correlations. Instead, window-level features are summarized at the recording level.

For feature x, the median is

$$x_{\text{median}} = Q_{50}, \tag{10}$$

and the interquartile range is

$$IQR = Q_{75} - Q_{25}. \tag{11}$$

Thus, each recording is represented by both its typical feature value and its variability across windows.

3.11 Machine Learning Pipeline

The model-processing pipeline contains:

1. Median imputation,
2. Variance-based filtering,
3. Correlation filtering,
4. ANOVA-based SelectKBest feature selection,
5. Classification.

Before training the classifiers, several preprocessing steps are applied to the 152 input features. Missing values are filled using the median of each feature. Features with zero or very low variance are removed, followed by correlation filtering with a threshold of 0.98 to reduce redundant information.

After these steps, the ANOVA F-score is used to rank the remaining features. The SelectKBest method retains the 15 highest-ranked features for classification.

Feature scaling is also applied to the models that require it. RobustScaler is used for Linear SVM, RBF SVM and Logistic Regression. Tree-based models, such as Random Forest and Extra Trees, are trained without this scaling step.

All preprocessing operations are fitted within the training portion of each cross-validation fold and then applied to its corresponding validation group.

Five classical classifiers were compared:

1. Random Forest,
2. Extra Trees,
3. Linear SVM,
4. RBF SVM,
5. Logistic Regression.

3.12 Group-Aware Validation

The dataset contains multiple recordings from the same physical bearing defect under different operating loads. For example, files with suffixes such as `_0`, `_1`, `_2` and `_3` represent different load conditions. Although these recordings are different, they originate from the same physical fault instance.

A conventional random train-test split could place recordings from the same defect in both training and validation sets. This may allow the model to learn characteristics of a defect that it has already encountered, resulting in an overly optimistic estimate of its performance.

To avoid this problem, Leave-One-Group-Out (LOGO) cross-validation is used. Recordings belonging to the same physical fault instance are assigned to one group. In each iteration, one complete group is kept aside for validation, while the model is trained using the remaining groups.

The final dataset contains 60 recordings from 15 physical fault-instance groups. Therefore, the LOGO procedure runs for 15 folds, with each group being used once as the validation group. This ensures that recordings from the same physical defect do not appear simultaneously in the training and validation sets.

The performance metrics are calculated using the combined out-of-fold predictions from all 15 folds. Since the same development groups were also used during model and feature selection, the resulting performance should be treated as a development cross-validation estimate rather than an independent final test result.

4. Results and Discussion

4.1 Model Comparison

The median-only representation was first used to compare the five classifiers.

Table 7: Classifier comparison using grouped cross-validation.

Model	Accuracy	Balanced Accuracy	Macro-F1
Extra Trees	78.33%	76.5%	75.4%
Random Forest	76.67%	74.4%	74.3%
Linear SVM	76.67%	74.4%	73.9%
RBF SVM	75.00%	72.3%	72.6%
Logistic Regression	73.33%	70.2%	69.9%

Extra Trees provides the strongest performance among the evaluated classifiers and is therefore selected for the subsequent comparison of feature representations.

4.2 Effect of IQR Features

The effect of adding window-level variability information was evaluated.

Table 8: Effect of median and IQR aggregation.

Representation	Accuracy	Balanced Accuracy	Macro-F1
Median-only	78.33%	76.5%	75.4%
Median + IQR	**83.33%**	**82.74%**	**82.0%**

The addition of IQR features improves the performance of the Extra Trees classifier. With median-only features, the model correctly classifies 47 out of 60 recordings. After adding IQR, the number of correct predictions increases to 50 out of 60.

The accuracy increases from 78.33% to 83.33%, giving an improvement of five percentage points. Balanced accuracy also increases from 76.5% to 82.74%, while macro-F1 improves from 75.4% to 82.0%.

This improvement suggests that the variation in feature values across different signal windows provides useful information in addition to their median values. However, these results are based on the same development groups used during model comparison and feature selection, so they should not be treated as independent external test performance.

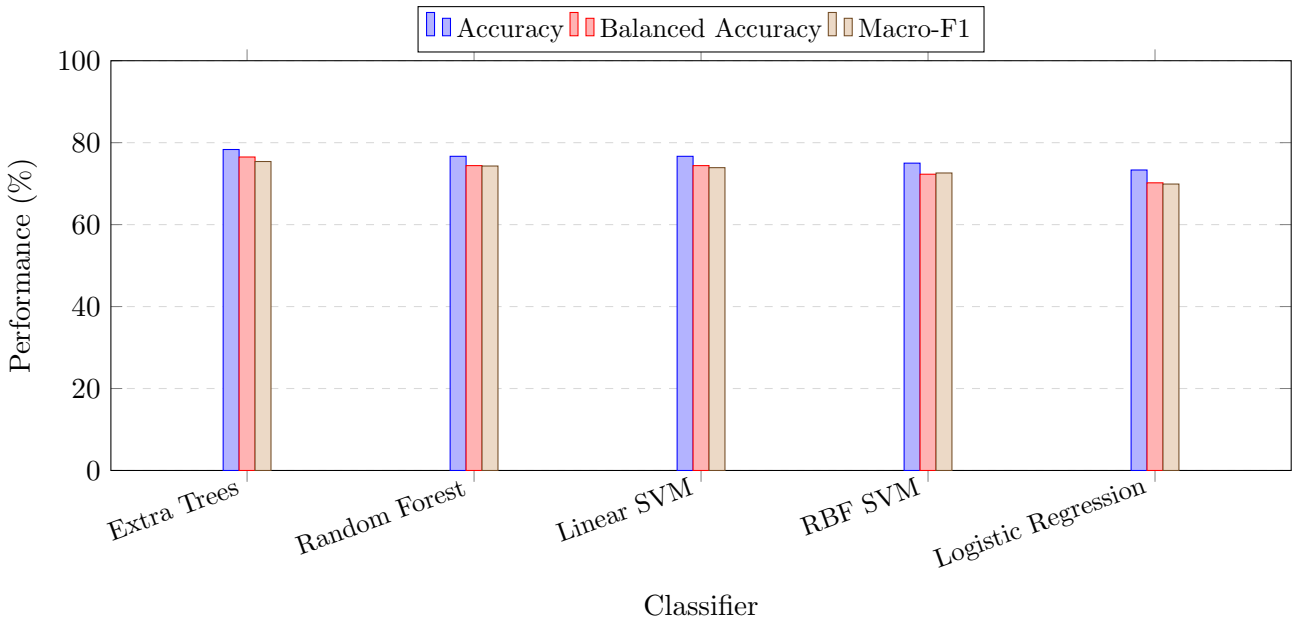


Figure 3: Comparison of five classifiers using median-only features under grouped cross-validation.

4.3 Class-Wise Performance

For the final median+IQR representation, the archived out-of-fold results show the following class-wise performance:

Table 9: Class-wise performance of the final development evaluation.

Class	Correct	Total	Accuracy
Ball fault	14	16	87.5%
Inner-race fault	12	16	75.0%
Outer-race fault	24	28	85.7%

The results show that the model performs differently across the three fault classes. Ball faults achieve an accuracy of 87.5%, while outer-race faults achieve 85.7%. Inner-race faults have the lowest accuracy at 75.0%, with 4 out of 16 recordings incorrectly classified as ball faults.

The confusion matrix also shows that 4 outer-race recordings were classified as ball faults, while 2 ball-fault recordings were classified as inner-race faults. Overall, 8 of the 10 incorrect predictions were assigned to the ball-fault class.

These results show why it is important to examine class-wise performance rather than relying only on overall accuracy. They also indicate that distinguishing some fault classes remains challenging for the current feature representation and classifier.

	Predicted Class		
Actual Class	**Ball**	**Inner-race**	**Outer-race**
Ball	**14**	2	0
Inner-race	4	**12**	0
Outer-race	4	0	**24**

Figure 4: Confusion matrix of the final Extra Trees classifier using median and IQR features. The entries represent pooled out-of-fold prediction counts from Leave-One-Group-Out cross-validation.

4.4 Performance Across Defect Sizes

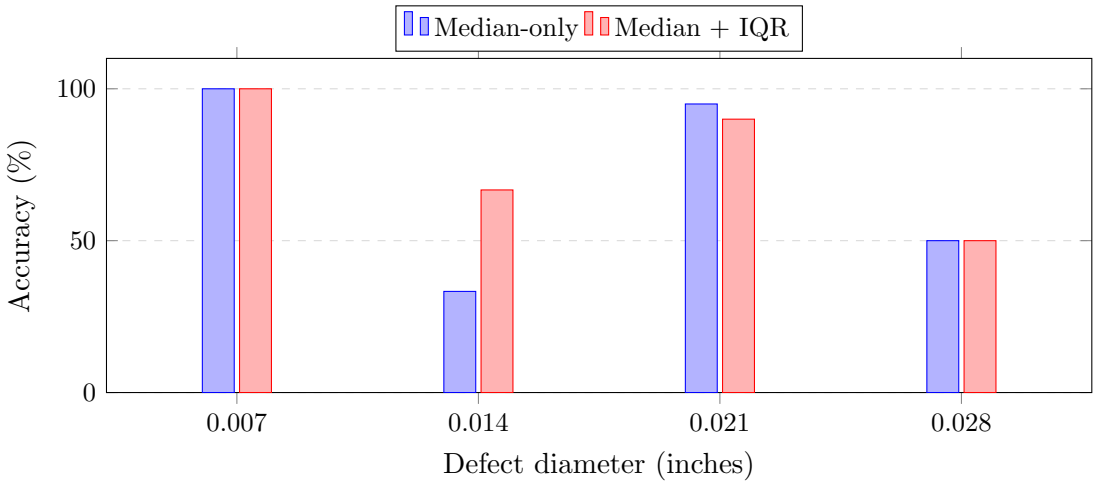


Figure 5: Comparison of classification accuracy across different bearing defect sizes.

The effect of adding IQR features varies across the different defect sizes. For the 0.007-inch subset, both feature representations correctly classify all 20 recordings. For the 0.014-inch subset, accuracy increases from 33.3% to 66.7% after adding IQR features, corresponding to an increase from 4 to 8 correct predictions out of 12 recordings.

For the 0.021-inch subset, accuracy decreases slightly from 95% to 90%, while both representations achieve 50% accuracy on the 0.028-inch subset. These results show that adding IQR features does not improve performance uniformly across all defect sizes.

The 0.028-inch subset contains only 8 recordings and has no outer-race examples. Therefore, its accuracy should be interpreted with this limited class coverage in mind.

4.5 Feature Distribution Across Bearing Fault Classes

To better understand the features used by the classifier, the distribution of one of the selected features was examined across the three bearing fault classes. The feature represents the envelope power ratio around the first inner-race fault harmonic in the 2000–3000 Hz frequency band.

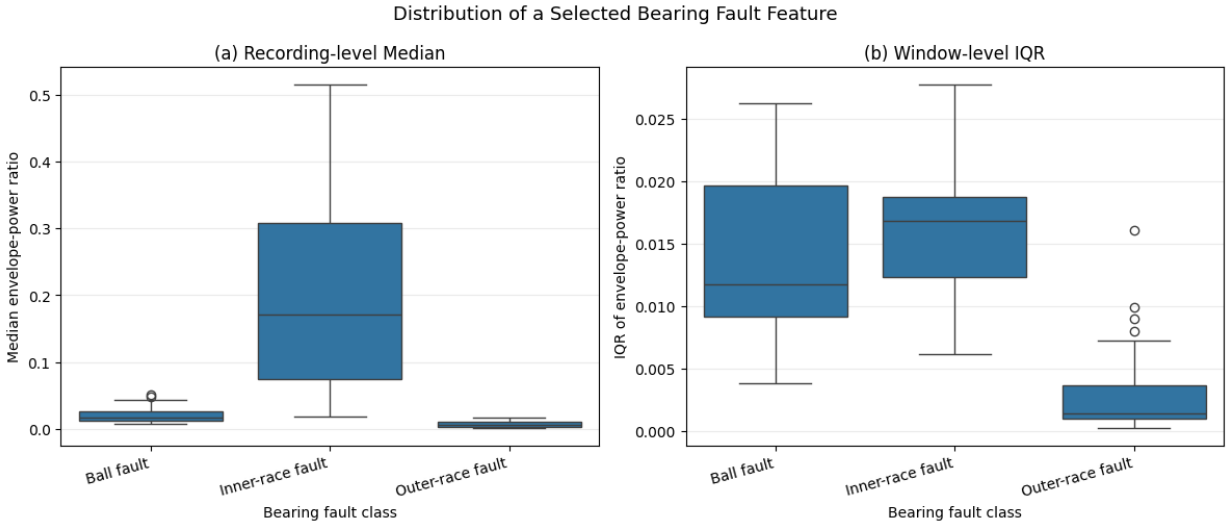


Figure 6: Distribution of a selected inner-race frequency feature across 60 recordings. The two panels show the recording-level median feature values and the corresponding window-level IQR values for each fault class.

Figure 6 compares the recording-level median and IQR values of the selected feature across the three fault classes. The median values are considerably higher for inner-race faults than for ball and outer-race faults. This shows that the selected frequency feature captures a useful difference in the vibration characteristics of inner-race faults.

The IQR plot also shows how much this feature varies across the signal windows of each recording. This variation provides additional information beyond the median value alone. However, the plot represents only one of the 15 features selected by the classifier. Its distribution should therefore be viewed as an illustration of the feature behaviour rather than proof that this feature can independently distinguish all three fault classes.

5. Challenges, Limitations and Discussion

5.1 Challenges Encountered

Several practical and methodological challenges were encountered during the development of the project.

Dataset organization: The project involved multiple feature tables and MATLAB recordings. Consistent normalization of filenames, labels and fault groups was necessary before model development.

Signal complexity: Raw vibration signals contain information from several mechanical components and do not necessarily expose the fault type directly. Multiband filtering and envelope analysis were therefore introduced.

Limited independent data: The final development set contains 60 recordings but only 15 physical fault groups. Consequently, the number of genuinely independent fault instances is considerably smaller than the number of recording files.

Information leakage: Different load recordings can originate from the same physical defect. This made conventional random cross-validation unsuitable as the primary evaluation strategy. Group-aware validation was therefore adopted.

Feature dimensionality: The feature extraction process produces a relatively large number of candidate features compared with the number of independent physical groups. Variance, correlation and statistical feature selection were used to obtain a compact representation.

5.2 Limitations

The current implementation has several limitations that should be considered while interpreting the results.

1. The classifier considers only three fault classes: ball fault, inner-race fault and outer-race fault. Healthy bearings are not included in the final classification model.
2. Although the development dataset contains 60 recordings, these belong to only 15 physical fault-instance groups. The number of independent fault instances is therefore limited.
3. The class distribution and defect-size combinations are not balanced. In particular, the 0.028-inch subset does not contain any outer-race examples.
4. The feature extractor is designed to generate 80 candidate features, but the final notebook retains only 76. The model uses median and IQR summaries of these features, followed by ANOVA-based selection of 15 features.
5. Model and feature selection were performed using the same development groups used for cross-validation. Repeated selection using the same groups may make the reported performance optimistic.
6. No completely untouched final test set was used for the reported 83.33% accuracy. The result should therefore be treated as a development cross-validation estimate.
7. Separate exploratory evaluations on JNU and UESTC showed limited transfer to other datasets. These experiments were supplementary and were not used to train the final CWRU model.
8. Prediction on a genuinely new MATLAB recording has not yet been verified through a completed inference run.

These limitations mean that the current system should be considered a research prototype rather than a general-purpose bearing fault diagnosis system.

6. Conclusion and Future Work

This project developed a machine learning-based approach for identifying bearing faults from vibration signals. The work combined signal processing, feature extraction and classical machine learning to distinguish between ball faults, inner-race faults and outer-race faults.

The vibration recordings were processed using a multiband Butterworth filter bank, Hilbert envelope analysis and Welch power spectral density estimation. Features were extracted from short signal windows and then summarized at the recording level using the median and interquartile range. Five classifiers were compared using Leave-One-Group-Out cross-validation, in which recordings from the same physical fault instance were kept in the same validation group.

Among the evaluated classifiers, Extra Trees gave the strongest results. The final representation using median and IQR features achieved 83.33% accuracy, 82.74% balanced accuracy and 82.0% macro-F1 across 60 recordings from 15 physical fault groups. The results also showed that the improvement from adding IQR features was not uniform across all defect sizes, and that inner-race faults were comparatively more difficult to classify.

Overall, the project demonstrates how signal processing and machine learning can be combined to identify different types of bearing faults. At the same time, the experiments highlight the need for careful validation and further testing before such a system can be used in practical condition-monitoring applications.

6.1 Future Work

There are several ways in which this work can be extended:

1. Include healthy-bearing recordings so that the system can distinguish normal operation from faulty conditions.
2. Expand the dataset with more independent physical fault instances and a better balance of fault classes and defect sizes.
3. Review the feature-selection step and evaluate the complete 80-feature representation alongside the current 76-feature representation.
4. Perform nested group-based validation or use a completely untouched group-level test set to obtain a more reliable estimate of model performance.
5. Test the complete inference pipeline on genuinely new MATLAB recordings and evaluate its performance on additional machines, sensors and operating conditions.
6. Explore more advanced machine learning and deep learning methods when a sufficiently large and diverse dataset is available.

References

[1] Case Western Reserve University, "Bearing Data Center." Available: https://engineering.case.edu/bearingdatacenter. Accessed: November 6, 2026.

[2] R. B. Randall and J. Antoni, "Rolling element bearing diagnostics—A tutorial," *Mechanical Systems and Signal Processing*, vol. 25, no. 2, pp. 485–520, 2011. doi: 10.1016/j.ymssp.2010.07.017.

[3] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ, USA: Pearson, 2010.

[4] P. D. Welch, "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," *IEEE Transactions on Audio and Electroacoustics*, vol. 15, no. 2, pp. 70–73, June 1967. doi: 10.1109/TAU.1967.1161901.

[5] P. Virtanen *et al.*, "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020. doi: 10.1038/s41592-019-0686-2.

[6] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, no. 85, pp. 2825–2830, 2011.

[7] P. Geurts, D. Ernst and L. Wehenkel, "Extremely randomized trees," *Machine Learning*, vol. 63, pp. 3–42, 2006. doi: 10.1007/s10994-006-6226-1.